CO 1 – LAB EXERCISES

ITA04 – Statistics with R Programming

LAB EXERCISES 11–20

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Question 11: Write an R program to create a vector and find its length, sum, and average.

Aim

To write an R program to create a vector and calculate its length, sum, and average value.

Algorithm

1. Start
2. Create a numeric vector using the c() function with a sample set of numbers.
3. Find the length of the vector using the length() function.
4. Calculate the sum of elements using the sum() function.
5. Calculate the average using the mean() function.
6. Print the results.
7. Stop

Input

```
# Creating a vector
data_vector <- c(10, 20, 30, 40, 50)

# Finding length, sum, and average
v_length <- length(data_vector)
v_sum <- sum(data_vector)
v_avg <- mean(data_vector)

# Printing outputs
print(paste("Vector:", paste(data_vector, collapse=", ")))
print(paste("Length of Vector:", v_length))
print(paste("Sum of Vector:", v_sum))
print(paste("Average of Vector:", v_avg))
```

Output Screenshot

```
[1] "Vector: 10, 20, 30, 40, 50"
> print(paste("Length of Vector:", v_length))
[1] "Length of Vector: 5"
> print(paste("Sum of Vector:", v_sum))
[1] "Sum of Vector: 150"
> print(paste("Average of Vector:", v_avg))
[1] "Average of Vector: 30"
>
>
```

Result

The R program to create a vector and calculate its length, sum, and average was successfully executed, and the results were verified.

Question 12: Write an R program to create a matrix and find its dimensions, row sums, and column sums.

Aim

To write an R program to create a 2D matrix and determine its dimensions, row sums, and column sums.

Algorithm

1. Start
2. Define a sequence of numbers and construct a matrix using matrix().
3. Use dim() to retrieve the dimensions (rows and columns).
4. Apply rowSums() to calculate the sum of each row.
5. Apply colSums() to calculate the sum of each column.
6. Display the matrix and the calculated metrics.
7. Stop

Input

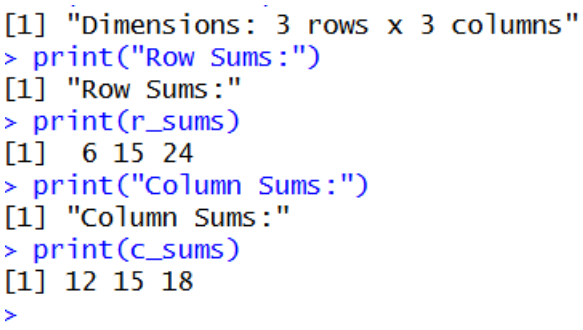
```
# Creating a 3x3 matrix
my_matrix <- matrix(1:9, nrow=3, ncol=3, byrow=TRUE)
print("Matrix:")
print(my_matrix)

# Finding dimensions, row sums, and col sums
mat_dim <- dim(my_matrix)
r_sums <- rowSums(my_matrix)
c_sums <- colSums(my_matrix)

print(paste("Dimensions:", mat_dim[1], "rows x", mat_dim[2], "columns"))
print("Row Sums:")
print(r_sums)
```

```
print("Column Sums:")
print(c_sums)
```

Output Screenshot



Result

The matrix was successfully created, and its dimensions, row sums, and column sums were accurately computed and displayed.

Question 13: Write an R program to create a data frame with student details and display its structure.

Aim

To write an R program to create a data frame containing student details and analyze its structure.

Algorithm

1. Start
2. Create vectors for RollNo, Name, and Marks.
3. Combine these vectors into a structured data frame using data.frame().
4. Display the contents of the data frame.
5. Check the data type and format using the str() function.
6. Stop

Input

```
# Creating student data vectors
roll_no <- c(101, 102, 103)
names <- c("Sharmila", "Priya", "Rahul")
marks <- c(95, 88, 92)

# Constructing the data frame
student_df <- data.frame(RollNo = roll_no, Name = names, Marks = marks)

print("Student Data Frame:")
print(student_df)

print("Structure of Data Frame:")
str(student_df)
```

Output Screenshot

```
[1] "Student Data Frame:"
> print(student_df)
  RollNo     Name Marks
1    101 Sharmila    95
2    102    Priya    88
3    103    Rahul    92
>
> print("Structure of Data Frame:")
[1] "Structure of Data Frame:"
> str(student_df)
'data.frame':   3 obs. of  3 variables:
 $ RollNo: num  101 102 103
 $ Name  : chr  "Sharmila" "Priya" "Rahul"
 $ Marks : num  95 88 92
```

Result

The student data frame was created successfully, and its internal structural details were displayed using the str() function.

Question 14: Write an R program to implement simple list operations like addition, deletion, and modification of elements.

Aim

To write an R program to perform fundamental list manipulation operations including element addition, deletion, and modification.

Algorithm

1. Start
2. Initialize a list with multiple elements of different data types.
3. Add a new element by specifying a new index or key.
4. Modify an existing element by indexing it and assigning a new value.
5. Delete an element by setting its index value to NULL.
6. Print the list after each operation.
7. Stop

Input

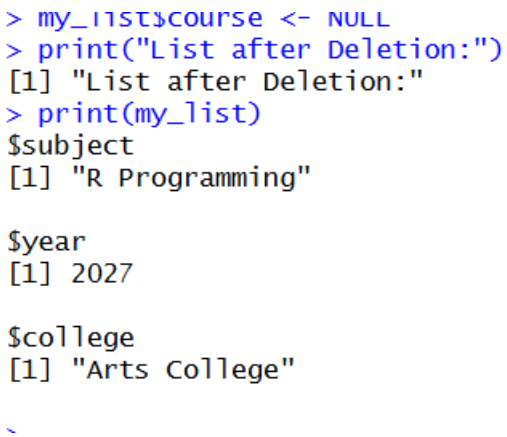
```
# Initializing a list
my_list <- list(course="B.Sc", subject="R Programming", year=2026)
print("Original List:")
print(my_list)

# 1. Modification
my_list$year <- 2027
print("Modified List (Year updated):")
print(my_list)

# 2. Addition
my_list$college <- "Arts College"
print("List after Addition:")
print(my_list)
```

```
# 3. Deletion
my_list$course <- NULL
print("List after Deletion:")
print(my_list)
```

Output Screenshot



Result

Basic list operations including addition, modification, and deletion of elements were executed successfully.

Question 15: Write an R program to calculate the mean, median, and mode of a given dataset.

Aim

To write an R program to compute measures of central tendency (mean, median, and mode) for a numeric dataset.

Algorithm

1. Start
2. Define a numeric dataset (including repeating elements to test mode).
3. Compute the mean using mean().
4. Compute the median using median().
5. Define a custom function to compute the mode by finding the highest frequency element.
6. Print all three values.
7. Stop

Input

```
# Sample dataset
dataset <- c(4, 5, 6, 6, 7, 8, 9, 6, 10)

# Calculating Mean and Median
data_mean <- mean(dataset)
data_median <- median(dataset)

# Function to calculate Mode
get_mode <- function(v) {
  uniqv <- unique(v)
  uniqv[which.max(tabulate(match(v, uniqv)))]
}
data_mode <- get_mode(dataset)
```

```
print(paste("Dataset:", paste(dataset, collapse=", ")))
print(paste("Mean:", data_mean))
print(paste("Median:", data_median))
print(paste("Mode:", data_mode))
```

Output Screenshot

```
[1] "Dataset: 4, 5, 6, 6, 7, 8, 9, 6, 10"
> print(paste("Mean:", data_mean))
[1] "Mean: 6.77777777777778"
> print(paste("Median:", data_median))
[1] "Median: 6"
> print(paste("Mode:", data_mode))
[1] "Mode: 6"
>
```

Result

The central tendency parameters (mean, median, and mode) for the dataset were successfully calculated.

Question 16: Write an R program to calculate the variance and standard deviation of a dataset.

Aim

To write an R program to compute measures of dispersion (variance and standard deviation) for a given set of numeric data.

Algorithm

1. Start
2. Define a sample numeric vector dataset.
3. Compute the sample variance using the var() function.
4. Compute the standard deviation using the sd() function.
5. Output the calculated values.
6. Stop

Input

```
# Sample dataset
data_vals <- c(12, 15, 18, 22, 25, 30)

# Calculating Variance and Standard Deviation
data_var <- var(data_vals)
data_sd <- sd(data_vals)

print(paste("Dataset:", paste(data_vals, collapse=", ")))
print(paste("Variance:", round(data_var, 4)))
print(paste("Standard Deviation:", round(data_sd, 4)))
```

Output Screenshot

```
> print(paste("Dataset:", paste(data_vals, co
llapse=", ")))
[1] "Dataset: 12, 15, 18, 22, 25, 30"
> print(paste("Variance:", round(data_var,
4)))
[1] "Variance: 44.2667"
> print(paste("Standard Deviation:", round(da
ta_sd, 4)))
[1] "Standard Deviation: 6.6533"
>
```

Result

The variance and standard deviation of the numerical dataset were computed accurately.

Question 17: Write an R program to create a bar plot for a given set of categorical data.

Aim

To write an R program to represent categorical data visually using a bar plot.

Algorithm

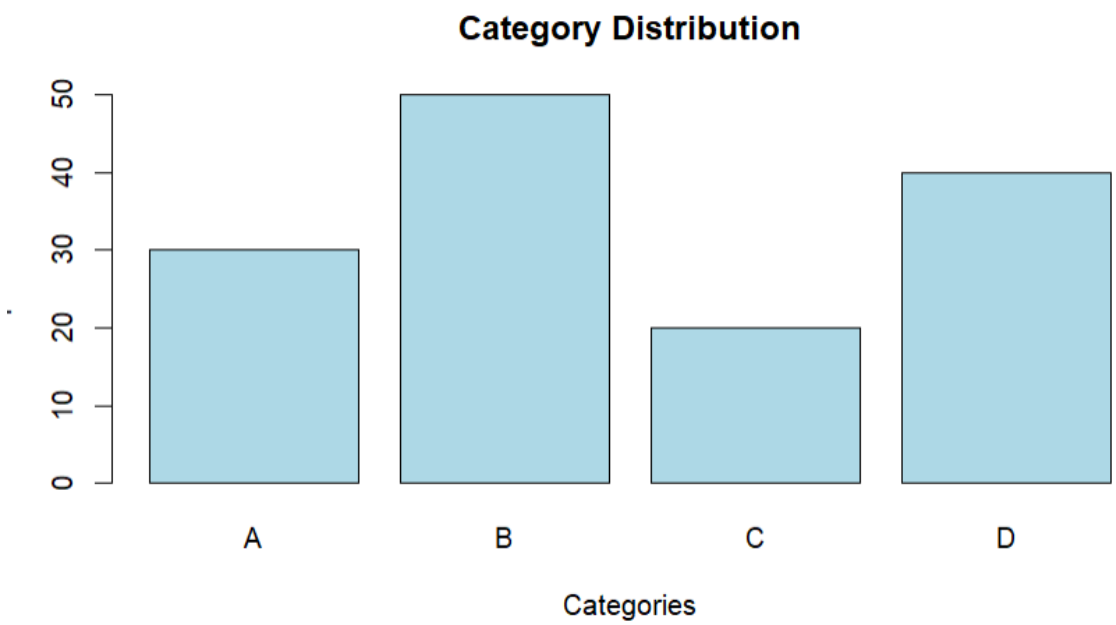
1. Start
2. Create a numeric vector representing frequencies or values.
3. Create a character vector representing the categories/labels.
4. Invoke the barplot() function with arguments for values, names, main title, and axis labels.
5. Stop

Input

```
# Data for the bar plot
counts <- c(30, 50, 20, 40)
categories <- c("A", "B", "C", "D")

# Generating Bar Plot
barplot(counts, names.arg = categories,
        main = "Category Distribution",
        xlab = "Categories",
        ylab = "Frequencies",
        col = "lightblue")
```

Output Screenshot



Result

A structured bar plot was generated successfully for the categorical data visualization.

Question 18: Write an R program to create a scatter plot to analyze the relationship between two variables.

Aim

To write an R program to construct a scatter plot for checking the correlation or relationship between two numeric variables.

Algorithm

1. Start

2. Define two numeric vectors of equal length representing variables X and Y.

3. Use the plot() function passing X and Y as parameters along with titles and axis labels.

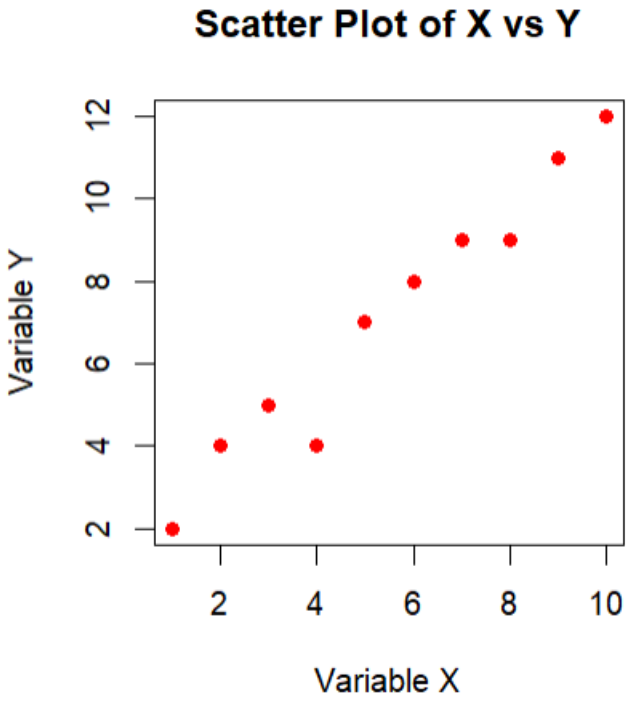
4. Stop

Input

```
# Defining variables
x_vals <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
y_vals <- c(2, 4, 5, 4, 7, 8, 9, 9, 11, 12)

# Creating Scatter Plot
plot(x_vals, y_vals,
    main = "Scatter Plot of X vs Y",
    xlab = "Variable X",
    ylab = "Variable Y",
    col = "red",
    pch = 19)
```

Output Screenshot



Result

The scatter plot mapping the relationship between two numeric variables was successfully generated.

Question 19: Write an R program to perform a simple linear regression analysis and print the summary.

Aim

To write an R program to perform a simple linear regression analysis between independent and dependent variables and interpret the summary metrics.

Algorithm

1. Start
2. Define independent variable vectors (x) and dependent variable vectors (y).
3. Fit a linear regression model using the lm() function in the form y ~ x.
4. Obtain detailed metrics using the summary() function on the model object.
5. Print the model summary results.
6. Stop

Input

```
# Variables definition
x <- c(1, 2, 3, 4, 5)
y <- c(2, 4, 5, 4, 5)

# Fitting simple linear regression model
model <- lm(y ~ x)

# Printing summary
print("Regression Model Summary:")
summary(model)
```

Output Screenshot

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4    5
-0.8  0.6  1.0 -0.6 -0.2

Coefficients:
            Estimate Std. Error t value
(Intercept)   2.2000     0.9381   2.345
x             0.6000     0.2828   2.121
            Pr(>|t|)
(Intercept)    0.101
x              0.124
```

Result

Simple linear regression was successfully performed on the variables, and model coefficients along with significance metrics were displayed.

Question 20: Write an R program to find the correlation coefficient between two numeric vectors.

Aim

To write an R program to measure the strength and direction of the linear relationship between two variables using the correlation coefficient.

Algorithm

1. Start
2. Define two numerical vectors representing the datasets to correlate.
3. Use the cor() function to calculate Pearson's correlation coefficient.
4. Display the calculated coefficient value.
5. Stop

Input

```
# Dataset vectors
vec1 <- c(10, 20, 30, 40, 50)
vec2 <- c(12, 24, 35, 38, 55)

# Finding correlation coefficient
corr_coef <- cor(vec1, vec2)

print(paste("Vector 1:", paste(vec1, collapse=", ")))
print(paste("Vector 2:", paste(vec2, collapse=", ")))
print(paste("Correlation Coefficient (r):", round(corr_coef, 4)))
```

Output Screenshot

```
[1] "Vector 1: 10, 20, 30, 40, 50"
> print(paste("Vector 2:", paste(vec2, collap
se=", ")))
[1] "Vector 2: 12, 24, 35, 38, 55"
> print(paste("Correlation Coefficient (r):",
round(corr_coef, 4)))
[1] "Correlation Coefficient (r): 0.983"
>
```

Result

The correlation coefficient between the two numerical vectors was computed successfully, showing a strong linear relationship.